For loop

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The Java for loop is used to iterate a part of the program several times. It is an advance form of while loop.

Syntax:

```
for(<any valid java statements>;<condition>;<any valid java statement>){
      //body of the loop
      …
      …
}
```

OR

famous in the books or internet

```
for(<initialization of variables>;<condition>;<update the loop control variable>){
      //body of the loop
      …
      …
}
```

we initialize loop control variable which is used in the condition.

```
for(<initialization of variables>;<condition>;<update the loop control variable>){
      //body of the loop
      …
      …
}
```

For example:

```
public class Q1a
{
    public static void main(String args[]){
        int start;
        System.out.print("\f");
        for(start = 1; start < 10; start++){
            System.out.print(start*start+", ");

        }

        System.out.print(start*start+"\n");
    }
}
```

False 5

1 2/ True 3

```
package Chapter4.forLoop;
import java.util.*;


/**
 * Write a description of class Q10 here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class Q10
{
    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
        System.out.print("\f");
        int num, revNum = 0, absDiff, rem, t;
```

```
        Scanner sc = new Scanner(System.in);
        System.out.print("\f");
        int num, revNum = 0, absDiff, rem, t;
        System.out.print("Enter any integer value: ");
        num = sc.nextInt();
```

It will execute every time whether the body of the for loop executes or not

```
        for(t = num; num > 0;){
            rem = num % 10;
            revNum = revNum*10 + rem;
            num /= 10;
        }
```

It will execute only once

```
        System.out.print("\nReverse: " + revNum);
        absDiff = Math.abs(t - revNum);
        System.out.print("\nAbsolute Difference = " + absDiff);
    }
}
```

optional

```
for(t = num; num > 0;){
```

① ② n

```
public class Q10
{
    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
        System.out.print("\f");
        int num, revNum = 0, absDiff, rem, t;
        System.out.print("Enter any integer value: ");
        num = sc.nextInt();
```

Snake loop.

F

```
        for(t = num; num > 0;rem = num % 10,revNum = revNum*10 + rem, num /= 10);
```

T

head

tail

```
        System.out.print("\nReverse: " + revNum);
        absDiff = Math.abs(t - revNum);
        System.out.print("\nAbsolute Difference = " + absDiff);
    }
}
```

optional

```
for(t = num; num > 0;){
```

① ② n

optional

for(① ; ② ; ③)

for(; ;)

if condition is not specified then by default it is true.

```
public class forLoopDemo
{
```

```
public class forLoopDemo
{
    public static void main(String args[]){
        System.out.print("\f");
        for(;;){
            System.out.print("\nFeeling infinite!!!!");
        }
    }
}
```

Exit condition is not specified, therefore this for loop will execute for infinite times.

```
public class forLoopDemo
{
    public static void main(String args[]){
        System.out.print("\f");
        int x = 0;
        for(;;){
            System.out.print("\nFeeling infinite!!!!");
            x++;
            if(x == 10){
                break;
            }
        }
        System.out.print("\nx = " + x);
    }
}
```

To make the loop finite, we can write exit condition in it

The break statement, terminates the current loop block.

```
package Chapter4.forLoop;


/**
 * Write a description of class forLoopDemo here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class forLoopDemo
{
    public static void main(String args[]){
        System.out.print("\f");
        int x = 0;
        for(System.out.print("\nFeeling infinite!!!!");;System.out.print("\nFeeling infinite!!!!")){

            x++;
            if(x == 10){
                break;
            }
        }
        System.out.print("\nx = " + x);
    }
}
```

1 N

we can write any valid Java statements inside the for loop ( ) brackets.